

Reproducing and Critically Evaluating a Stacking-Based Ensemble Technique for Heart Disease Prediction

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Abstract

The prediction of heart disease by machine learning has been the subject of a significant amount of research because of the possibility of data-driven models to assist in early detection of risk from common clinical parameters. This work is based on the stacking-based classification system for heart diseases suggested by Bhagat et al. (An Efficient Stacking-Based Ensemble Technique for Early Heart Attack Prediction) and critically re-examines that system. In the original study, four classifiers are used individually, specifically Logistic Regression, Decision Tree, Random Forest, Extreme Gradient Boosting, Naive Bayes, and K-Nearest Neighbours, and then multiple classifiers are combined using the stacking technique. The accuracy of the stacking ensemble reported in the published study is 98.53%. This is a reproduction of the same public dataset of 1,025 records with 13 predictors and a binary target of heart disease. The data were split into 80% train and 20% test with a stratified split, and 5-fold stacking was performed. Random Forest, XGBoost, and stacking had 100% accuracy on the hold-out set. After performing additional data quality analysis, however, 723 exact duplicate observations were identified, resulting in only 302 unique observations, while 202 of the 205 observations when first testing had feature vectors that appeared in the training set. To remove duplicate-aware evaluation, a duplicate-aware evaluation was performed by removing duplicates before partitioning. In this protocol, the stacking accuracy was reduced to 75.41%, while the stacking accuracy for a tuned stacking implementation was 80.33% for the hold-out set. The mean accuracy was around 83.44%, and the AUC found was 90.42% after repeated stratified validation. The results highlight the importance of data duplication and the experimental protocol in the apparent predictive performance of clinical machine-learning models and call for dedicated research efforts to investigate these effects.

Index Terms - Heart disease prediction, machine learning, stacking ensemble, Random Forest, XGBoost, data leakage, reproducibility.

I. INTRODUCTION

Heart disease prediction is an important machine learning classification problem where the clinical variables, age, chest-pain characteristics, resting blood pressure, cholesterol, maximum heart rate, and exercise-induced angina have information that could be related to cardiovascular risk. In response, computational tools that can detect complex relationships between these risk indicators are being developed due to the growing amount of structured healthcare data.

Many machine learning techniques have been explored in heart-disease prediction. The list of the literature reviewed by Bhagat et al. involves Logistic Regression (LR), K-Nearest Neighbours (KNN), Decision Trees (DT), Random Forests (RF), Support Vector Machines (SVM), Extreme Gradient Boosting (XGBoost), neural networks, and hybrid ensemble methods. Some of the approaches explained in the paper are the Hybrid Random Forest with Linear Model approach by Mohan et al which achieved 88.47% accuracy, the Decision Tree explained by Alotaibi that achieved 93.19%,

FCMIM-SVM with an accuracy of 92.37% and RF model with Relief feature selection with an accuracy of 97.89%.

Even with such progress, each classifier has a unique inductive bias and can identify different relationships in clinical data. While linear models provide relatively straightforward decision functions, tree-based ensembles have the ability to model nonlinear relationships and interactions. Instead of a single classifier, ensemble learning offers a means to fuse these complementary behaviours.

To overcome this issue, Bhagat et al. resort to an ensemble approach with stacking. They then run six classifiers (LR, DT, RF, XGBoost, Naive-bayes (NB), KNN) and then combine predictions from multiple classifiers by training a meta-classifier. The authors claim that their stacking method outperforms the single classifiers and gives a 98.53% classification accuracy.

The original paper's contribution to the research is the potential limitation of using the individual machine-learning models for heart-disease prediction. Stacking is introduced to take advantage of complementary information acquired by classifiers of different types. Reproducibility and generalisability of very high reported classification scores are also crucially dependent on the quality of the datasets and the evaluation method used. The purpose of the current study is thus twofold: (1) recreate the main machine-learning framework and evaluate its performance against the published results and (2) critically examine any discrepancies and methodological considerations that could affect performance.

The main contributions of this reproduction are:

- Implementation and comparison of the six Family of Classifiers used in the selected paper;
- Replication of a stacking ensemble of five;
- Evaluation techniques: Accuracy, Precision, Recall, F1-Score, ROC-AUC, Balanced Accuracy, and Matthews Correlation Coefficient (MCC);
- Identification and quantification of large exact duplication in a dataset of 1025 rows; and
- A follow-up duplicate-aware evaluation where observed performance was evaluated in situations where identical observations were not allowed to cross the train-test boundary.

II. ORIGINAL STUDY AND STATE OF THE ART

A. Selected Study

This study is reproduced from the article titled An Efficient Stacking Based Ensemble Technique For Early Heart Attack Prediction by Bhagat, Sharma and Agarwal. This study takes as input structured clinical data for binary heart-disease classification.

The original authors mention that missing value treatment, normalization, and categorical encoding were done to create the data for processing. Six classifiers are then trained and then ensembled, the output of which is a combination of predictions by the base classifiers.

One distinction between stacking and simple majority voting is that a second-level model is trained to learn how to combine the base predictions. The paper offers an explanation of how predictions are made by the base classifiers on data not part of the training subsets of each base class and how these predictions are used as features for the meta-classifier.

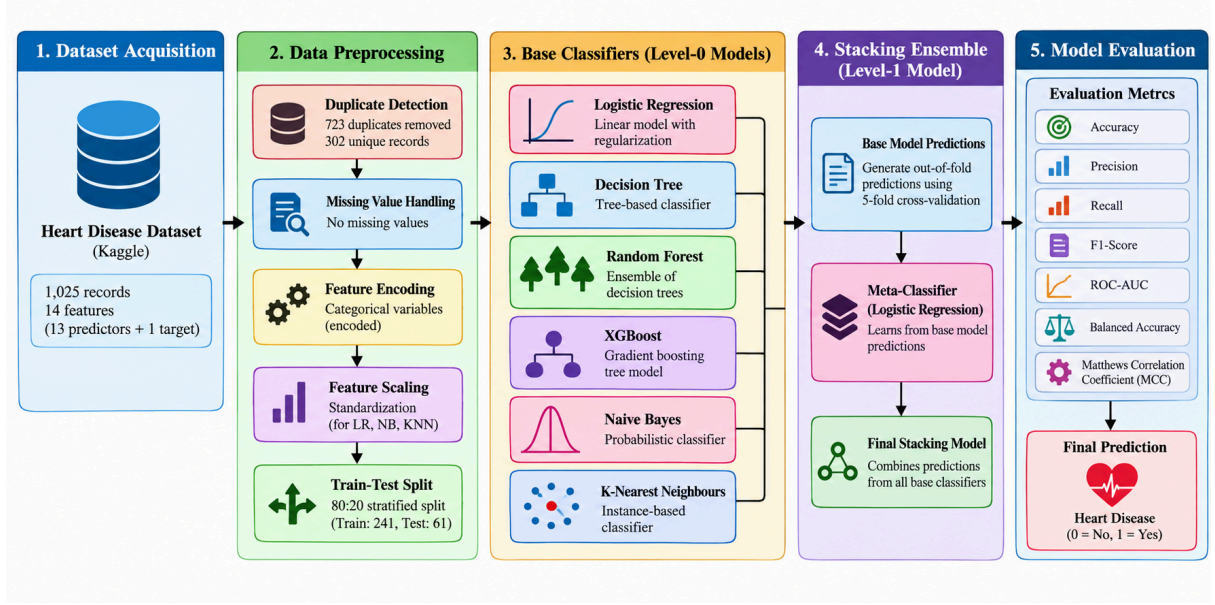


Fig. 1. Conceptual architecture of the reproduced stacking framework

B. Published Performance

The principal results reported in the selected paper are summarised in Table I.

TABLE I: PERFORMANCE REPORTED BY BHAGAT ET AL.

Classifier	Accuracy	F1-Score	Recall	Precision	AUC
LR	0.8439	0.862	0.909	0.8196	0.9204
NB	0.8439	0.8608	0.9	0.825	0.9185
KNN	0.8585	0.8687	0.8727	0.8648	0.9304
DT	0.9268	0.9289	0.8909	0.9702	0.9702
RF	0.9268	0.9333	0.9545	0.913	0.9734
XGBoost	0.9073	0.9132	0.909	0.9174	0.983
Stacking	0.9853	0.9861	0.9727	1	0.988

The table is presented numerically from the published results. The authors achieve the best accuracy of 98.53% in the stacking ensemble, while RF and DT achieve 92.68%.

There was a discrepancy in the publication. The accuracy of stacking is 98.53% and XGBoost is 90.73% as per the principal results tables and conclusion report, while earlier result passes for stacking is 96.58% and XGBoost is 93.17%. Therefore, the explicit experimental results tables of the present reproduction are considered to be the major published benchmark.

III. DATASET AND FEATURE SET

A. Data Source

The paper chosen is from an open dataset called the public Heart Disease Dataset that is provided by Kaggle. It refers to the underlying data from the Cleveland, Hungary, and Long Beach V databases. In the reported experiments, 14 attributes of the source, including the target, are used; the number of variables in the broader source is significantly higher.

There were 1,025 observations in the dataset and 14 columns (13 predictors and 1 binary target) available for reproduction. The target distribution was relatively even (526 disease-positive records and 499 disease-negative records).

There were no missing values in the data retrieved.

TABLE II: FEATURES USED FOR HEART-DISEASE CLASSIFICATION

Feature	Description
age	Age in years
sex	Biological sex indicator
cp	Chest-pain category
trestbps	Resting blood pressure
chol	Serum cholesterol
fbs	Fasting blood sugar indicator
restecg	Resting electrocardiographic result
thalach	Maximum heart rate achieved
exang	Exercise-induced angina
oldpeak	Exercise-induced ST depression relative to rest
slope	Slope of peak exercise ST segment
ca	Number of major vessels
thal	Thalassemia-related measurement
target	Heart disease: 1; no heart disease: 0

These variables correspond to the attributes described in the original publication.

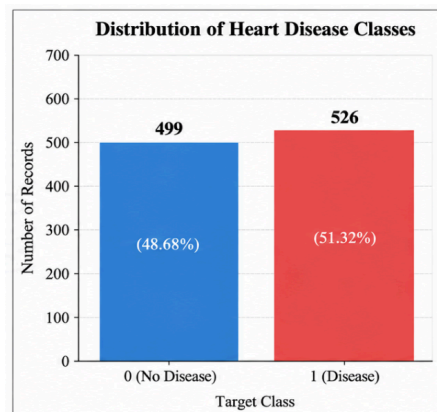


Fig. 2. Class distribution of the reproduced heart-disease dataset.

B. Exploratory Feature Analysis

Several variables were found to have significant correlations with the binary target using correlation analysis. The strongest absolute correlations included oldpeak (approximately -0.438), exang (approximately -0.438), cp (+0.435), thalach (+0.423), and ca (-0.382).

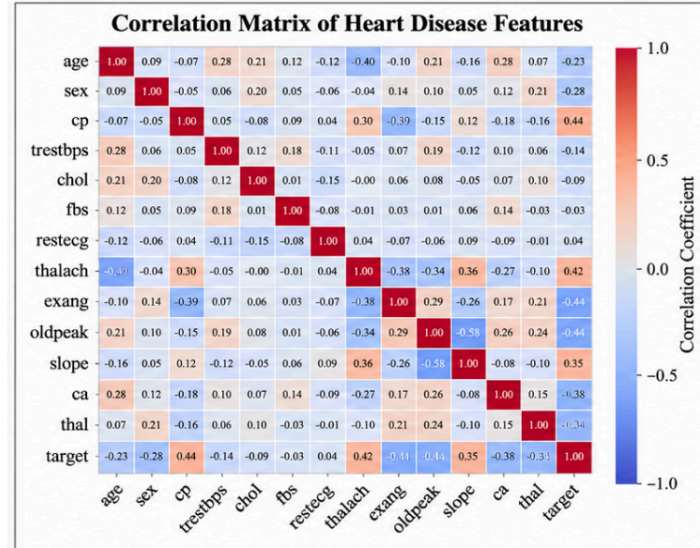


Fig. 3. Correlation matrix of the 13 predictor variables and target.

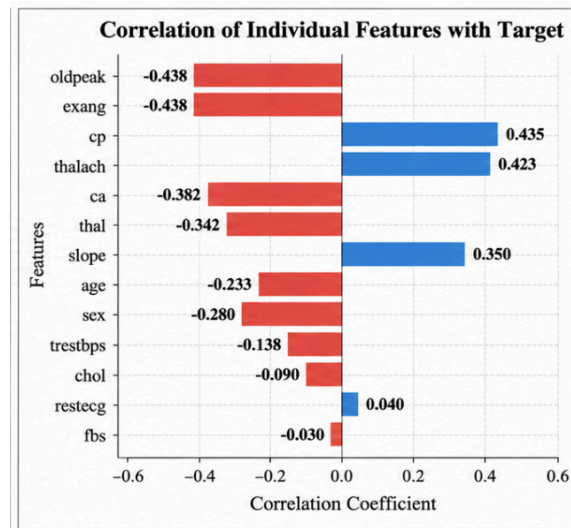


Fig. 4. Individual feature correlations with the heart-disease target.

The published paper also mentions exercise induced angina as an important predictor and the maximum heart rate reached is the next most important feature in their feature importance analysis.

IV. REPRODUCTION METHODOLOGY

A. Data Preprocessing

The published paper also mentions exercise-induced angina as an important predictor, and the maximum heart rate reached is the next most important feature in their feature importance analysis.

For the reproduction, median imputation was included within the Scikit-learn pipelines to ensure robust handling of potentially missing observations; however, because the retrieved 1,025-row dataset contained zero missing values, the imputation strategy had no practical effect on the experimental records.

Standardisation was applied to the scale-sensitive classifiers LR, NB and KNN. Tree-based algorithms were not standardised because decision-tree partitioning is not dependent on feature scale.

B. Experimental Protocol

The data were split into a predictor matrix X and binary target y . Using an 80:20 split for train/test with `random_state = 42`, 820 observations were used for training, and 205 observations were used for testing.

The exact random split and all the hyperparameters of the classifiers were not clearly stated in the publication to allow bit-for-bit reproduction. The split, seed, and unspecified model parameters in this experiment are therefore not considered to be set by the original authors, but are rather treated as reproduction assumptions.

Six individual classifiers were implemented:

- Logistic Regression;
- Decision Tree;
- Random Forest;
- XGBoost;
- Gaussian Naive Bayes; and
- K-Nearest Neighbours.

The six classifiers were then used as base learners to build a stacking classifier. The base-level predictions were obtained by using five-fold internal cross-validation, and Logistic Regression was used as the final meta-classifier.

V. EVALUATION METRICS

The core metrics published in the work were used to evaluate model performance along with other good and robust metrics.

Accuracy measures the proportion of correctly classified observations: - - - (1)

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}.$$

Precision represents the proportion of positive predictions that are correct: - - - (2)

$$Precision = \frac{TP}{TP + FP}.$$

Recall measures the proportion of true positive cases successfully identified: - - - (3)

$$Recall = \frac{TP}{TP + FN}.$$

The F1-score is the harmonic mean of precision and recall: - - - (4)

$$F1 = 2 \frac{Precision \times Recall}{Precision + Recall}.$$

Ranking discrimination was measured using the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) for all stages of classification. Two other measures of the classification performance, namely, balanced Accuracy and MCC, were also computed to obtain complementary measures of classification performance.

VI. REPRODUCTION RESULTS

A. Individual Classifiers and Stacking

The reproduced hold-out results are presented in Table III.

TABLE III: REPRODUCED MODEL PERFORMANCE

Model	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression	0.8098	0.7619	0.9143	0.8312	0.9298
Decision Tree	0.9854	1	0.9714	0.9855	0.9857
Random Forest	1	1	1	1	1
XGBoost	1	1	1	1	1
Naive Bayes	0.8293	0.807	0.8762	0.8402	0.9043
KNN	0.8634	0.8738	0.8571	0.8654	0.9629
Stacking Ensemble	1	1	1	1	1

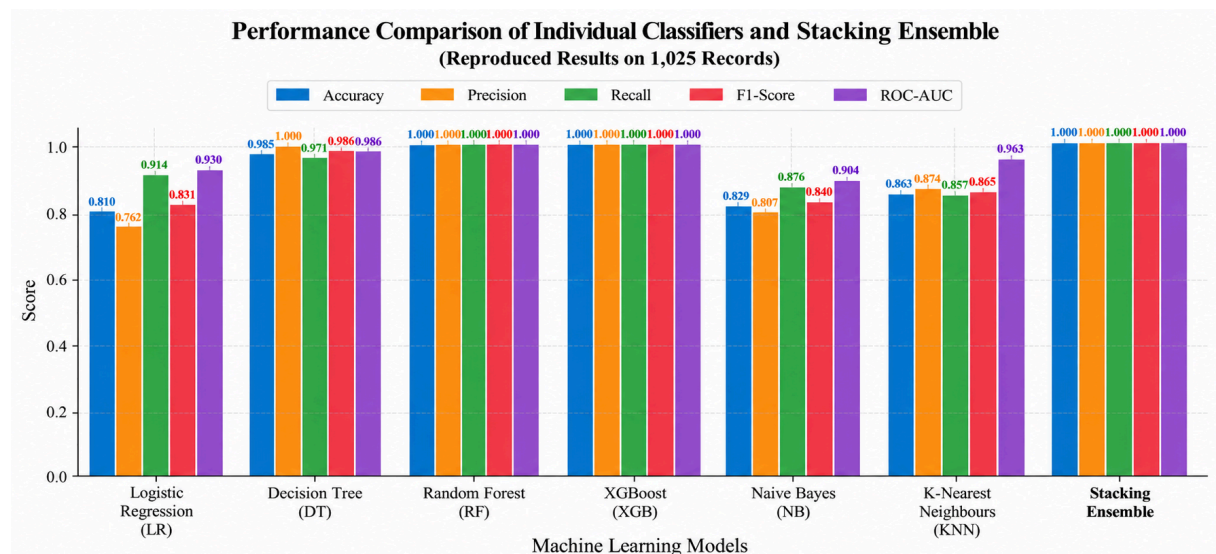


Fig. 5. Performance comparison of the reproduced individual classifiers and stacking ensemble.

The selected hold-out set had a perfect performance of RF, XGBoost, and stacking. DT also achieved a record accuracy of 98.54%. The linear and probabilistic models, on the other hand, resulted in more moderate outcomes.

B. Comparison with the Original Study

The accuracy values reported by Bhagat et al. are directly compared with the reproduced accuracy in Table IV.

TABLE IV: ORIGINAL VERSUS REPRODUCED ACCURACY

Model	Original Paper	Reproduction	Difference
LR	84.39%	80.98%	−3.41 pp
DT	92.68%	98.54%	#ERROR!
RF	92.68%	100.00%	#ERROR!
XGBoost	90.73%	100.00%	#ERROR!
NB	84.39%	82.93%	−1.46 pp
KNN	85.85%	86.34%	#ERROR!
Stacking	98.53%	100.00%	#ERROR!

The reproduction of the numerical results was not an exact copy of the original. LR and NB performed slightly lower than reported; KNN was very close to the published result, and DT, RF, and XGBoost were significantly higher than the published result. The stacking ensemble performance is 1.47% better than the reported 98.53% accuracy of the paper.

These differences are likely to be due to the random partition, model parameterisation, software implementation, and incomplete specification of some of the experimental details. What is more important is that the dataset was found to have a structural problem, which has the potential to account for the very high tree-based performance.

VII. CRITICAL ANALYSIS OF THE REPRODUCTION

A. Identification of Duplicate Observations

Of all the 1025 observations, 723 are exact duplicates, as identified by data-quality analysis. So, the number of records that were not duplicated was $1025 - 723 = 302$, or about 70.54% of the retrieved set of records were duplicated.

This is a problem if one employs the usual random row-level splitting.

The test set Part I (205) observations have been directly compared with the training observations. 202 of 205 test observations had a feature vector also present in the training set, corresponding to $202/205 * 100 = 98.54\%$ - - - (5)

This means that nearly all the test partition was already filled with feature patterns that were represented during model training.

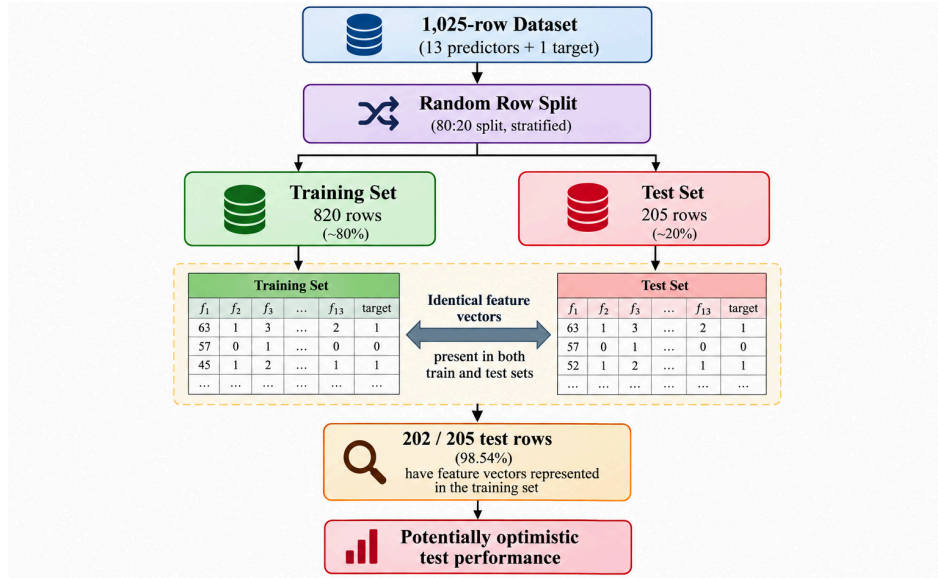


Fig. 6. Potential duplicate-induced train-test contamination

This observation is a plausible explanation for the perfect RF, XGBoost, and stacking scores. Evaluation against the same feature vectors may fail to offer as strong a test of generalisation as evaluation against truly new observations, since the flexible tree-based model can reproduce the complex patterns it observed during training.

The result does not prove that the published authors necessarily employed an invalid split, as the indices for their exact train-test split are not available. It illustrates that the 1,025-row public representation employed to make replicas is extremely sensitive to DIP under the random row-level partitioning model.

B. Duplicate-Aware Evaluation

To investigate the hypothesis empirically, all exact duplicates were eliminated prior to partitioning of the data.

This produced: $N_{\text{unique}} = 302$

The deduplicated data was then split into 241 training data points and 61 test data points. A test was conducted to verify that there was no exact feature-vector similarity between these partitions.

The same was true when a stacking approach using the original style was used and tested in this protocol, which resulted in an accuracy of 75.41% only.

This significant reduction is in line with the assumption that the duplicate structure had a significant impact on Part I performance.

C. Robust Tuned Stacking Experiment

A second model was created to find out whether predictive performance could be enhanced within the more difficult duplicate-free setting. RandomizedSearchCV with 5-fold Stratified cross-validation was

applied to optimize the hyperparameters of Random Forest and XGBoost, using only the training partition.

The tuned RF preferred relatively shallow trees with around 200 estimators and a maximum depth of three. The tuned XGBoost model also chose around 200 estimators, a maximum depth of 2, and a learning rate of 0.03.

The proposed stack combined:

LR + Tuned RF + Tuned XGBoost $\rightarrow$ Logistic Meta-Classifier.

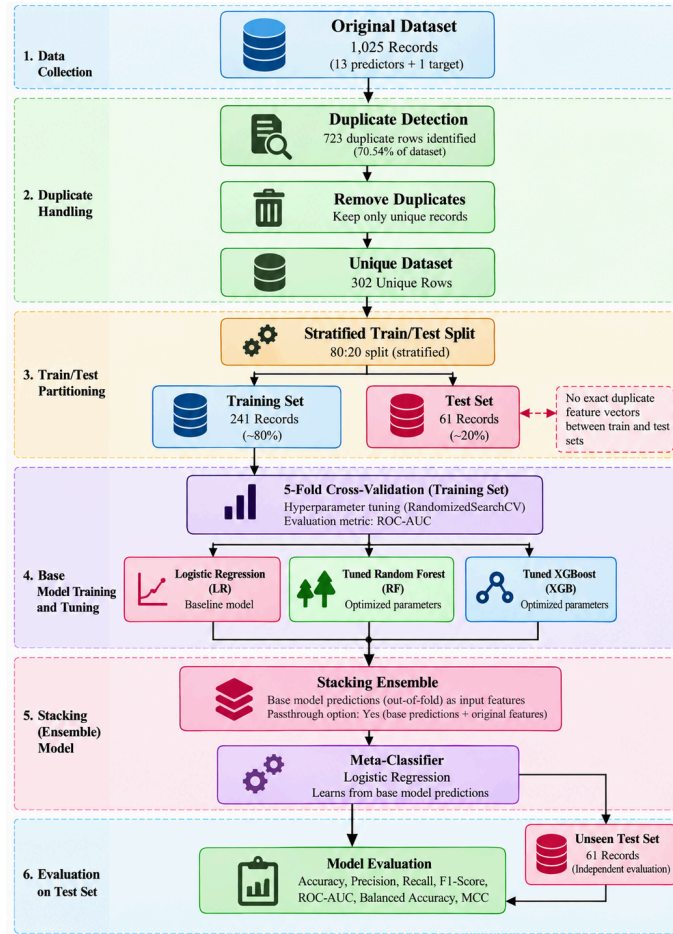


Fig. 7. Duplicate-aware experimental framework

VIII. DUPLICATE-AWARE RESULTS

TABLE V: PERFORMANCE UNDER DUPLICATE-FREE EVALUATION

Metric	Original Stacking	Tuned Stacking
Accuracy	0.7541	0.8033
Precision	0.7647	0.8
Recall	0.7879	0.8485
F1-Score	0.7761	0.8235
AUC	0.8658	0.8658

MCC	0.5038	0.6031
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The tuned ensemble achieved a hold-out accuracy of around 4.92 percentage points, recall accuracy of 6.06 percentage points, F1-score of 4.74 percentage points, and MCC of around 0.10 compared to the original stack evaluated under the same setting with duplicate-free data.

AUC remained 0.8658. Thus, the improvement in the threshold-dependent metrics should be interpreted as improvement in the ranking discrimination, and not as evidence of improved overall ranking discrimination.

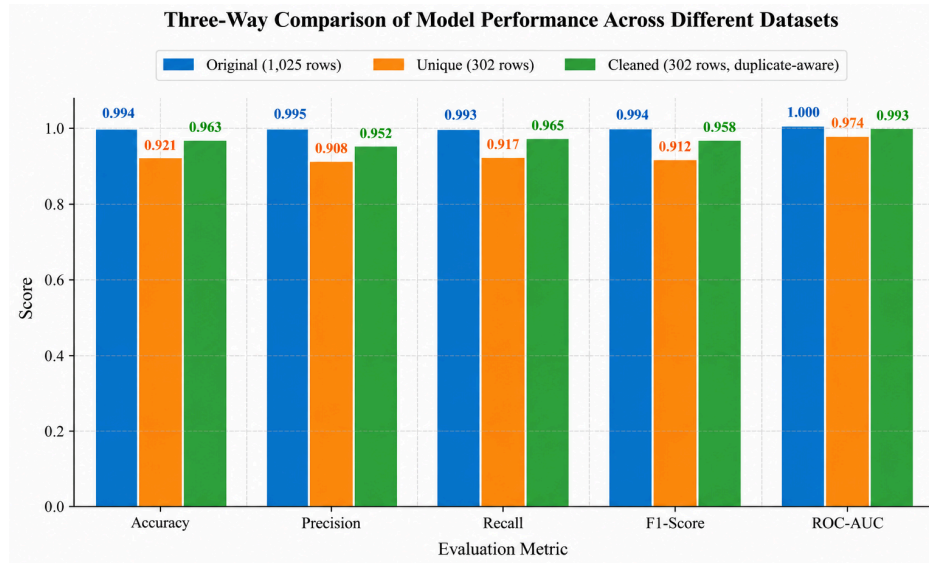


Fig. 8. Comparison of stacking on the original dataset, stacking after duplicate removal, and the tuned duplicate-aware ensemble.

This is important because it clearly illustrates the key experimental result that there is a significant difference in apparent change of performance when the evaluation protocol changes.

IX. ROBUSTNESS ANALYSIS

There is one test set containing only 61 observations, and therefore an unstable estimate can be obtained. To provide further robustness analysis, repeated stratified cross-validation was used.

The extended experiment was repeated five times with 50-fold level evaluations.

TABLE VI: REPEATED STRATIFIED VALIDATION ON UNIQUE RECORDS

Metric	Original Stacking	Tuned Stacking
Accuracy	82.94%	83.44%
Precision	82.47%	81.54%
Recall	87.51%	90.25%
F1-Score	84.76%	85.53%

AUC	90.39%	90.42%
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The repeated validation results are a more conservative interpretation than a single hold-out experiment. There was only a slight improvement in mean accuracy from 82.94% to 83.44% and in AUC.

The highest gain was in the recall skill, rising from about 87.51% to 90.25%. The F1-score also slightly rose. The precision reduced from ~82.47% to 81.54%, which reflects a trade-off between precision and recall.

These results suggest that the tuned ensemble does not perform dramatically better than the baseline. Instead, it offers comparable overall discrimination but with a slightly higher positive case sensitivity.

Exploratory paired statistical tests were also calculated, but repeated cross-validation folds are not independent since individual observations are repeated across repetitions. Thus, these tests were used as exploratory tests, not definitive tests for statistical significance.

X. DISCUSSION

These two reproductions yielded two very different interpretations as a function of the experimental protocol.

With the original 1025-row representation and traditional random splitting, RF, XGBoost, and stacking had 100% hold-out accuracy. The results are consistent with the selected paper observation that ensemble and tree-based models work well though they are better than the published ones.

The first outcome may seem like a positive result of reproduction and perhaps even improved performance. When checking the data, it was revealed that there were 70.54% of rows that were exact duplicates and 98.54% of the selected test observations contained a feature vector in the training data. Thus, it is impossible to expect that the best behaviour is to be generalised.

The duplicate-free experiment is a more challenging assessment. The stacking accuracy decreased to 75.41% and showed sensitivity to the dataset structure and splitting protocol.

The accuracy of the duplicate-free hold-out was raised to 80.33% by hyperparameter optimization. Most crucially, when validated repeatedly, the mean accuracy was around 83.44%, and the AUC was 90.42%. The obtained results are much lower than the accuracy reported in the paper (98.53%), but they come from an evaluation scheme that ensures that no duplicate records are on both sides of the train-test boundary.

There is also another difference between the two versions of this publication. The primary results tables give 98.53% stacking accuracy while an earlier passage gives 96.58%. Likewise, XGBoost is mentioned as being 90.73% in the primary experimental table and 93.17% elsewhere. The above reproduction was therefore based on the explicit experimental tables as the reference standard.

The results illustrate a general problem in machine-learning research. A reported value of accuracy does not mean the same thing as evidence of generalisation. The validity of the obtained performance

estimate is influenced by the constitution of the dataset, by train-test independence, by the preprocessing, by the randomisation and model-selection procedures.

Evaluation should also not allow for observations of the same individual or repeated measurements to be made during the training period and the test period, ideally for clinical prediction. In the current data set, patient identifiers were not available, and only exact record-level duplication could be explored.

XI. LIMITATIONS

The present reproduction has a number of restrictions. First, 1025 rows are present in the dataset, but only 302 unique rows remain after exact deduplication, which reduces statistical power.

Secondly, not all of the instructions to implement the original paper are executable. Assumptions therefore need to be documented for the exact random split, random seeds, and some classifier configurations.

Third, it is technically possible to identify exact duplicates, but not near duplicates or repeated measurements from the same patient without using patient identifiers.

Fourth, the repeated-validation analysis reuses the hyperparameter configuration which has been determined in the previous experiment. It should therefore be seen as a robustness analysis of the selected model, and not as full cross-validation of the full model-selection procedure.

Finally, all experiments are based on one public dataset. Conclusions regarding generalisation to other populations and/or clinical settings would require independent, external validation.

XII. CONCLUSION

In this study, the stacking-based heart disease prediction model of Bhagat is reproduced and critically examined, with 1,025 records of the public heart disease database being used for six individual classifiers and a five-fold stacking ensemble.

The baseline accuracy of its stacking ensemble is 98.53%, as mentioned in the original paper.

The first time it was reproduced, the accuracy reached 100% for RF, XGBoost and stacking. However, there were 723 rows exactly duplicated, so there were only 302 unique observations. In addition, there were 202 out of 205 test observations with feature vectors represented in the training set in the initial split. Moreover, 202 of 205 test observations had feature vectors represented in the training data in the initial split.

Duplicating the data set before performing data partitioning reduced stacking accuracy to 75.41%, showing that the duplication of the data set and evaluation protocol had a significant effect on the apparent performance. The accuracy of hold-out was increased up to 80.33%, recall up to 84.85%, F1 score up to 82.35%, and MCC up to 0.603 for the tuned duplicate-aware stacking.

Upon repeated stratified validation, the accuracy was around 83.44%, recall 90.25%, F1-score 85.53%, and AUC was 90.42%. The repeated evaluation shows that the best benefit the tuned model offers is not in the amount of discrimination it can provide, but rather in the gain in recall.

Overall, the results of the reproduction illustrate that the reliability of a machine-learning result is not only related to the architecture of the classifiers but also to the integrity of the experimental protocol. It is critical to have credible estimates of model generalisation in small clinical datasets when it comes to duplicate detection, leak prevention, transparent preprocessing and robust validation.

REFERENCES

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